Lab experiment-15

Decision Tree using Python

Aim

To write a Python program to implement a Decision Tree classification algorithm.

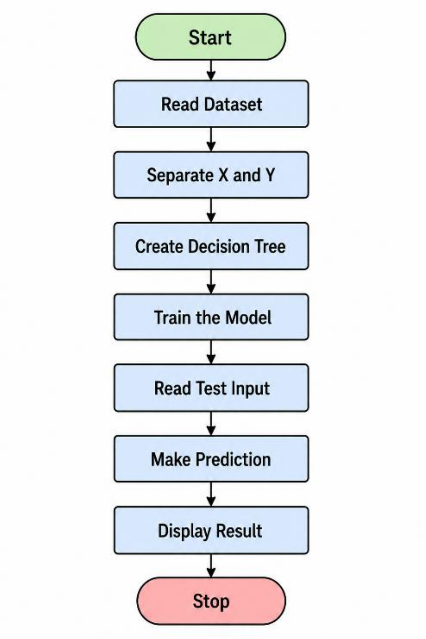
Objective

To build a Decision Tree model.

Algorithm

1. Start.
2. Read the dataset.
3. Separate input features and target values.
4. Create a Decision Tree classifier.
5. Train the model using the dataset.
6. Read test input from the user.
7. Predict the class.
8. Display the result.
9. Stop.

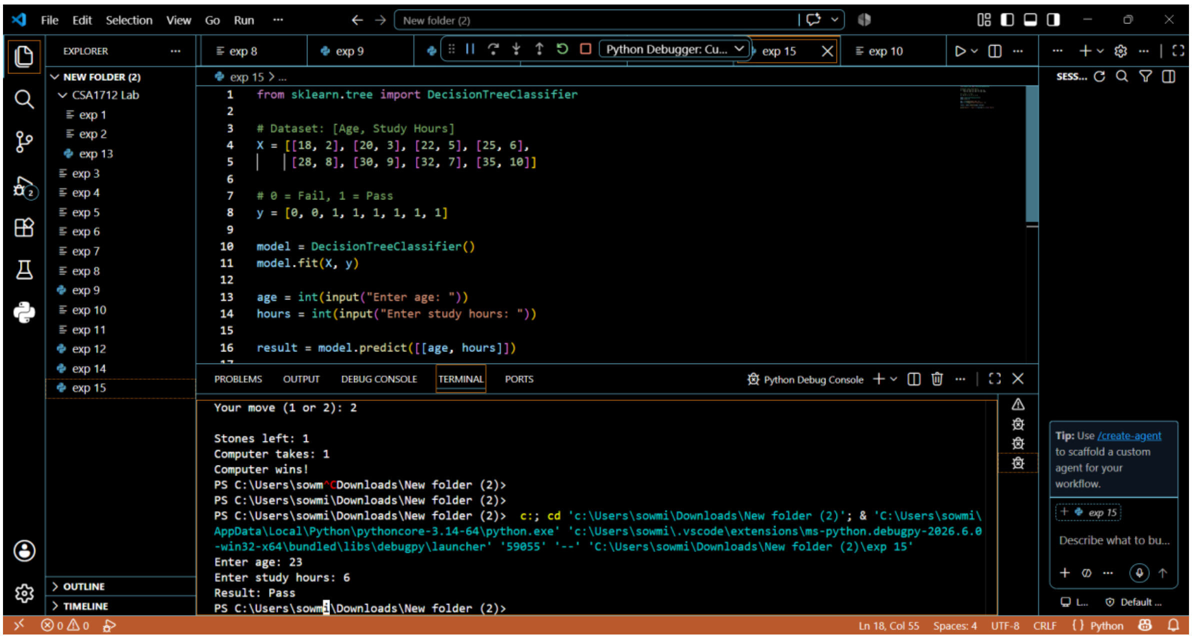
Flowchart



Code

```
from sklearn.tree import DecisionTreeClassifier
# Dataset: [Age, Study Hours]
X = [[18, 2], [20, 3], [22, 5], [25, 6],
     [28, 8], [30, 9], [32, 7], [35, 10]]
# 0 = Fail, 1 = Pass
y = [0, 0, 1, 1, 1, 1, 1, 1]
model = DecisionTreeClassifier()
model.fit(X, y)
age = int(input("Enter age: "))
hours = int(input("Enter study hours: "))
result = model.predict([[age, hours]])
print("Result:", "Pass" if result[0] == 1 else "Fail")
```

Output



Result
The Decision Tree classifier was implemented successfully.

Conclusion
The program successfully classifies the input data using a Decision Tree.